

1). What is Linux?

-> Linux is an open-source Operating System that manages a computer's hardware and software resources.

-> Linux was Developed by Linus Torvalds, in 1991.

-> It is used by Globally across Servers,cloud infrastructure,embedded system and desktop computers.

-> Linux is highly stable and secure,customizable and provides strong support for Software Development and Network based Applications.

2). What is the difference between Hard Link & Soft Link?

-> Hard Link :

Hard link is additional name for Existing File.It points directly to the same inode.

Hard link cannot span across different file system and cannot be created for directories.

If the original file is deleted, the data is still accessible through the hard link because the inode remains untill all links are removed.

Eg: In file1.txt file1_hard.txt

ls -li file1.txt file1_hard.txt

-> Soft Link:

Soft Link is a special file that points to the path of another file.

Soft link can span across different file systems and can link to directories.

If the Original file is deleted,the soft link becomes broken link,since it only stored the path.

Eg: ln -s file1.txt file1_soft.txt

ls -li file1.txt file1_soft.txt

3). What is Kernel in linux?

-> In linux, Kernel is the Fundamental core of Operating System acting as intermediary between computer's Hardware and Software components.

-> It manages the resources linke the cpu, memory and device handling communication between them and ensuring the all application can run efficiently without conflicts.

4). How do you create a user account?

-> In linux, by using the "useradd" command to create a new user.

-> In terminal, sudo useradd user_name

5). What is the 'grep' command used for in Linux?

-> In linux, grep command is used to search for specific pattern or text within the files.

-> To filter the text based keyword and search inside the files.

6). Step1: Create user p1

Step2: He should be part of 3 groups g1,g2,g3.

Step3: whenever he creates a file automatically in the group section of file grp g1 should come.

-> Answer:

```
dharani-gilli@dhara...$ sudo useradd p1
dharani-gilli@dhara...$ sudo passwd p1
New password:
BAD PASSWORD: The password fails the dictionary check - it is based on a dictionary word
Retype new password:
Sorry, passwords do not match.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
dharani-gilli@dhara...$ sudo groupadd g1
dharani-gilli@dhara...$ sudo groupadd g2
dharani-gilli@dhara...$ sudo groupadd g3
dharani-gilli@dhara...$ sudo usermod -aG g1,g2,g3 p1
sudo: usermod: command not found
dharani-gilli@dhara...$ sudo usermod -aG g1,g2,g3 p1
dharani-gilli@dhara...$ sudo usermod -g g1 p1
dharani-gilli@dhara...$
```

7. Step1: Create directory /tmp/bg as root user and create files inside it.

Step2: “abhi” should be the owner of the directory. He should be able to create files and delete files inside the directory and also he should be able to add content to all files inside the directory.

->Answer :

```
dharani-gilli@dhara...$ sudo useradd abhi
dharani-gilli@dhara...$ sudo passwd abhi
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
dharani-gilli@dhara...$ mkdir /tmp/bg
mkdir: cannot create directory '/tmp/bg': File exists
```

```
dharani-gilli@dhara...$ sudo passwd abhi
New password:
Retype new password:
passwd: password updated successfully
dharani-gilli@dhara...$ su - abhi
Password:
$ cd /tmp/bg
$ touch text.txt
$ echo "Hi I am Dharani kumar" > text.txt
$ cat text.txt
Hi I am Dharani kumar
$ ls -l
total 4
-rw-rw-r-- 1 abhi abhi 0 Sep 18 11:42 file1
-rw-rw-r-- 1 abhi abhi 0 Sep 18 11:42 file2
-rw-rw-r-- 1 abhi abhi 22 Sep 18 11:48 text.txt
$ ^C
$ ls -ld /tmp/bg
drwxrwx--- 2 abhi abhi 4096 Sep 18 11:48 /tmp/bg
$ exist
-sh: 7: exist: not found
$
dharani-gilli@dhara...$
```

8). You suspect that a particular process is consuming excessive CPU resources on your Linux server. How would you identify and terminate this process?

-> In linux, we identify the excessive CPU resources by using the command of top , htop which displays the running process along with their CPU usage.

-> From the result We identify the Process Id(PID) which consumes high CPU consumption.

-> To Terminate we use the kill command [kill (PID)].

-> Eg: kill PID.